MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU –B.VOC. (Honours)
Regular Examination November 2025
First Semester
(2025 ADMISSION ONWARDS)
Skill Development Components (SDC) - MG1SDCFTA100 – FUNDAMENTALS OF FOOD TECHNOLOGY

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Part A
One Sentence
Answer any 10
Each question carries 1 mark

1. The net weight is the weight of a product without its container or package.
2. 1gm of Carbohydrate = 4 Kcal
3. Emotional needs – feeling of security, love, attention
4. Saccharomyces boulardii, bifidobacterium bifidum, lactobacillus acidophilus,streptococcus salivarius
5. Disaccharides – sucrose, lactose, maltose
6. Microminerals – Iron, Zinc, Copper, Iodine etc
7. EAA–Phenylalnine,Valine,Threonine,Tryptophan,methionine,Leucine,Isoleucine,Lysine, Histidine
8. The Food and Agriculture Organization (FAO) defines prebiotics as "non-viable food components, non-digestible fibers, that provide health benefits to the host by modulating the microbiota (They selectively promote the growth and/or activity of beneficial bacteria, such as Lactobacilli and Bifidobacteria, in the gut)"
9. Anti nutritional factors - enzyme inhibitors, such as trypsin inhibitor, chymotrypsin inhibitor and α-amylase inhibitor, lectins, tannins, phytic acid, oxalates, phenolic compounds, saponins and oligosaccharides (flatulence factors)
10. Benefits of food fortification - improve the nutritional quality of the food supply and to provide a public health benefit with minimal risk to health
11. Less treatment time, low treatment temperature, substitute for conventional heat pasteurization, Improve yield etc.
12. Mixing Equipments – Tumbling mixers, Ribbon mixers

Part B
Short Answer
Answer any 4 questions
Each question carries 5 marks

13. Responsible for ensuring that the quality policies adopted by a company are followed. It helps to identify and prepare the necessary SOPs related to the control of quality. It must determine that the product meets all the applicable specifications and that it was manufactured according to the internal standards of GMP. QA also holds responsibility for

quality monitoring or audit function. QA functions to assess operations continually and to advise and guide them towards full compliance with all applicable internal and external regulations.

14. The acidity of food is its **concentration of hydrogen ions** measured on the pH scale, where a lower pH indicates higher acidity. Acidity plays a crucial role in food processing by acting as a preservative, enhancing flavor, improving texture, and maintaining consistency.
15. Size is an important attribute of foods used for screening solids to separate foreign materials & grading of fruits & vegetables. In fluid flow, heat & mass transfer calculations, it is necessary to know the size of the sample.

    Example: Smaller particles may impact pumping. & mixing process (as the particles becomes more viscous) In the case of large particles, may occur sedimentation/separation. Size of the particulate food is also critical.

    Particle size of powdered milk must be large enough to prevent agglomeration but small enough to allow rapid dissolution during reconstitution.

    Particle sizes are expressed in different units coarse particles measured in (mm), fine particles are measured in terms of screen/ mesh size, very fine particles in (nm)
16. Functions of Vitamin A
    a. Prevents night blindness
    b. Maintains healthy epithelical cells – Protective lining of internal and external body surfaces
    c. Essential for growth and development of skeltal and soft tissues
    d. Formation and maintenance of tooth enamel and healthy gums
17. Limitations of organic farming - lower yields, higher production costs, and greater labor intensity for tasks like pest and weed control
18. Sulphur dioxide has been used in foods for long and as a general preservative. It is used in the treatment of fruits and vegetables before and after dehydration to extend the storage life, prevent the growth of undesirable microorganisms and in making of fruit juices. It is also the most useful agent for the preservation of browning reactions is dried fruits. Most cut fruits are treated with sulphur dioxide to prevent enzymatic browning

Part C
Essay
Answer ant 2
Each question carries 10 marks

19. Career opportunities in food technology are varied and span from product development and quality control to research, safety, and management roles across different industries. Explain
20. With examples

Saturated Fats	Unsaturated Fats
Contains a single bond.	Contains at least one double bond.
Not to be consumed more than 10 percent of total calories per day.	Not to be consumed more than 30 percent of total calories per day.
Excessive consumption leads to heart diseases.	Good for consumption, but excessive may increase cholesterol.
Increases low- density lipoproteins (LDL), which is called as bad cholesterol.	Increases High-density lipoprotein (HDL), which is commonly known as good cholesterol and also reduce low-density lipoproteins (LDL).
Would not spoil quickly.	Spoil quickly.
Foods sources of saturated fats are whole milk, butter, cheese, margarine, coconut oil, vegetable oil, meat, peanut, fried foods, etc.	Foods sources of unsaturated fats are walnuts, flax, avocado, sunflower oil, soybean oil, fish oil, canola oil, red meat, etc.
High melting point.	Low melting point.
Solid state in room temperature.	Liquid state in room temperature.

21. Explain pasteurization with types - Low Temperature Holding or LTH), High-Temperature Short-Time (HTST), and Ultra-High-Temperature (UHT)**. Include at least two**
22. Drying process and factors affecting drying rate - Factors affecting the rate of drying include environmental conditions like air temperature, humidity, and velocity; material properties such as surface area, thickness, and initial moisture content etc.